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Design and Analysis of a Tensegrity-Based Tumbleweed Robot with Controlled Locomotion

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***Abstract*—The robotic systems designed based on the Tensegrity has benefits of lightweight construction, Most existing designs are based on structural compliance and resilience, but do not make use of structural compliance. continuous internal actuation and consequently consume high energy. This paper will present the design and analysis of a tumbleweed robot based on tensegrity. Focusing on the principles that rely on minimal actuation and produce locomotion mainly by wind blowing through the environment. A Nonlinear Model Predictive Control externally, (NMPC) framework guides the direction of motion with the help of cables. An Unscented Kalman Filter (UKF) is used for state estimation, and an actuator is used to control the system. The simulations indicate that the locomotion is sustained at around 40 m for 30 s.The simulations demonstrate continuous locomotion at an average of 40 m for 30 s. A high level of displacement, an average velocity of 1.35 m/s and low control energy (≈107 J) Stress is structurally determined and is within safe limits and restrictions, validating an energy-assisted locomotion paradigm in which the robot is pushed by environmental forces with control being applied selectively for directional guidance.**

***Keywords*—Tensegrity, Tumbleweed Robot, Wind-driven locomotion, Nonlinear model. Predictive control, Unscented Kalman filter, Energy-assisted locomotion..**

I. Introduction

A. Background and Motivation

Plants have evolved some passive mobility mechanisms such as the tumbleweed, detaching from roots and rolling across terrain under the forces of the wind to cover large areas with minimal energy. This behavior motivates a class of tumbleweed-inspired robots that use passive or semi-passive rolling to high area coverage and low mechanical complexity [12], [16]. Such designs Unlike conventional wind pumps, they can be built very simply and use less energy. That's the problem with wheeled and legged systems that have rigid architectures, impact High control complexity, and vulnerability. Passive tumbleweed systems, but they are not directionally controllable, which is why there is a need for selective control in the form of actuation which maintains structural efficiency and allows for intentional control and guidance.

Tensegrity structures offer a potential structure for this purpose. A tensegrity system is a combination of rigid bars, connected to one another by pre-tensioned cables, creating a balance of tension compression for shape retention and stability [1]. The tensile forces are applied throughout the structure to create an open lattice framework. They remain elastic when loaded and return to their original shape when unloaded force networks. Compliance, shock absorption and resilience [2] are offered as damage to the structure. It is these properties which make them acceptable for deformation-driven locomotion‘11 they can be selectively adjusted to the particular conditions. The center of mass changes, causing the cable lengths to shift, and eliminating uncontrolled rolling. The only constraint imposed are the rigid joints or direct limb actuation [5], [8].

B. Challenges and Research Gaps

There are many open challenges in the area of tensegrity tumbleweed robots. The dynamics have complex interactions between each other they are nonlinear, coupled and underactuated. Modelling and control is challenging with cables, bars and external forces. Existing methods are based on simplified dynamic models or heuristic models. Strategies without the cable elasticity, actuator saturation, and contact effects [7]. Autonomous navigation strategies combining reinforcement in learning and sensor fusion, similar problems have been tackled in neither a comprehensive framework nor extensive publications exist for collaborative robot systems incorporating

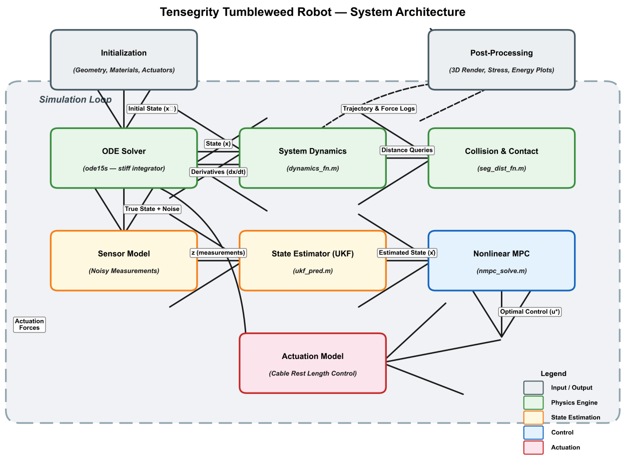

Fig. 1. Closed loop interaction between components of the system in the system architecture When the UKF is included, the sensor measurements and the structural dynamics plant are among the structural dynamics. The state estimator, NMPC controller and actuator module.

high fidelity dynamics, high degree of control. Tensegrity tumbleweed locomotion has not yet been shown to be feasible with estimation. [14]

C. Proposed Work and Contributions

The design and analysis of a tensegrity based tumbleweed is presented here. A wind powered robot that moves its body using interaction and selective internal cable actuation only for directional guidance. A TT-6 tensegrity architecture is used in the proposed system it is the six-bar, twelve-node structure with thirty cables been supplied and contains a detailed dynamic model which accounts for the elasticity of the cables, the rigidity of the bars and the dynamics of the actuators, etc. An NMPC framework regulates the cable , UKF gives a very good state estimation in a finite prediction horizon, and the actuation. The measurement noise and model uncertainty are estimated [6], [7], [10]. The trajectory tracking, energy saving performance over 30 s in numerical simulation is evaluated. A new benchmark is set for structural integrity, and estimation performance establishing a basis for future experimental validation [13].

II. Methodology

This system architecture brings and integrates structural dynamics, sensing, estimation, control, and actuation in a unified closed-loop environment, as shown in the Fig. 1. Initialization establishes geometric configuration, material-properties, prestress-conditions, and actuator parameters, followed sequentially by dynamic modeling, state estimation, and control computation.

The sensor measurements from the sensors were shown in Fig. 1 as shown physical plant are fed to the UKF leading to filtered estimates of Nodal positions and nodal velocities. Such estimates are provided to the NMPC, which chooses the optimal action at time t from a set of actions and rewards that depend on the current state of the system. outputs cable-length commands.Those commands are used by the actuator module closing the feedback loop to the tensegrity structure. This modular uncertainty of estimation and the design of the control are separated in a clean manner in the architecture, and Enable independence of tuning and validation for each component .

A. Dynamic Modeling

Every node has forces applied to it from the tension of the cables, compression of the bars, and gravity, interaction with wind [3], [4]. Newton's Law of Motion is used for node i. second law:

$$\ddot{\mathbf{q}}_i = \frac{\mathbf{F}_i}{m_{\text{node}}} \tag{1}$$

where the total nodal force is:

$$\mathbf{F}_i = \mathbf{F}_{\text{cable},i} + \mathbf{F}_{\text{bar},i} + \mathbf{F}_{\text{grav},i} + \mathbf{F}_{\text{contact},i} + \mathbf{F}_{\text{fric},i} + \mathbf{F}_{\text{wind},i} + \mathbf{F}_{\text{damp},i} \tag{2}$$

Rigid bars are simulated by bilateral penalty springs. Ground contact uses takes into account penalty normal force and Coulomb and viscous friction. Wind loading applies a quadratic drag force to nodes above the structure's center of Mass (COM) represents a kind of level-0 cover and is the only variable used to approximate actual ground level cover [12]. The damping is proportional to the velocity of each node, and it is applied to all nodes. The equations and the integration of motion is done with a stiff numerical integrator (`ode15s`).

Cable tension uses a softplus activation for smooth differentiability:

$$\sigma(s) = \tfrac{1}{2}\left(s + \sqrt{s^2 + \varepsilon}\right), \quad f_{\text{cable}} = \max\!\big(0,\ k_{\text{cable}}\,\sigma(s) + u_{m,i}\big) \tag{3}$$

where $s = L - L_0$ is the cable elongation and $u_{m,i}$ is the elongation of the cable and $u_{m,i}$ is the actuator contribution. This formulation is different from a hard in that there is no issue of non-differentiability. The point is that there is a max(0, s) clamp that will slow down the gradient-based optimization.

B. State Estimation and Control

An Unscented Kalman Filter (UKF) is used for noisy sensor observations which are able to deal with nonlinear dynamics and uncertainty in the system [7] sending accurate state estimates to controller. This is consistent arm systems in the field of adaptive estimation.Collaborative Robotic Arm systems with adaptive estimation approaches were shown in the field of adaptive estimation. systems [15], [17].

The NMPC framework gets the UKF state estimates and optimizes a finite horizon N_p cost function [6], [10]:

$$J = \sum_{k=1}^{N_p} \left[w_k\, \mathbf{e}_k^\top Q\, \mathbf{e}_k + \mathbf{u}_k^\top R\, \mathbf{u}_k + \Delta\mathbf{u}_k^\top R_d\, \Delta\mathbf{u}_k \right] \tag{4}$$

where $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}_{\text{ref}}$ state tracking error, $\Delta\mathbf{u}_k = \mathbf{u}_k - \mathbf{u}_{k-1}$ is the control increment and the terminal weight is $w_{N_p} = 3$ (else 1). A reduced order linear prediction model is used to

TABLE I
SIMULATION PARAMETERS

Parameter	Symbol	Value
Structure		
Nodes / Bars / Cables	N, N_b, N_c	12, 6, 30
Node mass	m_{node}	0.12 kg
Robot radius	r_{robot}	≈0.95 m
Stiffness and Damping		
Cable stiffness	k_c	2000 N/m
Bar stiffness	k_b	20000 N/m
Ground spring	k_g	5000 N/m
Ground dashpot	d_g	150 N·s/m
Structural damping	c_d	0.5 N·s/m
Bar damping	c_b	5.0 N·s/m
Contact and Friction		
Coulomb friction coefficient	μ_c	0.3
Viscous friction coefficient	μ_v	0.5
Ground slope	θ	0.03 rad
Motor Model		
Natural frequency	ω_n	25 rad/s
Damping ratio	ζ	0.7
Max actuator force	$u_{\max}$	40 N
Min cable length	$L_{\min}$	0.50 m
Max cable length	$L_{\max}$	1.40 m
Aerodynamics		
Wind velocity vector	$\mathbf{v}_{\text{wind}}$	[8.0, 1.0, 0] m/s
Drag coefficient	C_d	1.0
Air density	ρ	1.225 kg/m^3
Drag area per node	A_{node}	0.07 m^2
NMPC		
Prediction horizon	N_p	6
Control limit	u_{lim}	0.15 m
Rate limit	Δu_{lim}	0.06 m
Prediction time step	Δt_p	0.05 s
Integration		
Time step	Δt	0.05 s
Simulation duration	T	30.0 s
ODE solver	—	`ode15s` (stiff)

model the dominant translational modes. The full dynamics of the rolling COM is used for computational tractability. The optimization subject to the box constraints:

$$\|\mathbf{u}_k\|_\infty \le u_{\text{lim}}, \qquad \|\Delta\mathbf{u}_k\|_\infty \le \Delta u_{\text{lim}} \tag{5}$$

A full list of simulation parameters can be found in Table I. The table is organized in six groups of parameters. The *Structure* group defines the *TT-6* topology (12 nodes, 6 bars, 30 cables), the mass of the nodes and the overall robot radius. The *Stiffness and Damping* group defines separate spring constants for cables and bars, and ground contact spring and dashpot values in the penalty contact model. The *Contact and Friction* group provides the Coulomb and viscous friction coefficients and the ground slope angle. Second-order natural frequency, damping ratio, maximum output force of actuators, and limits on cable lengths are in the group of *Motor Model*. The *Aerodynamics* group specifies the applied wind velocity vector, drag coefficient, air density, and effective drag area per exposed node.Finally, the *NMPC* group states the prediction horizon length, control and rate limits, and the prediction time step used by the optimizer.

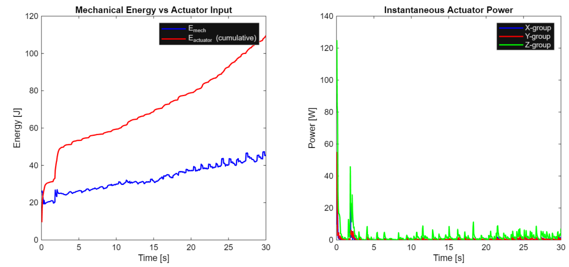

Fig. 2. Energy budget and actuation profile over 30 s. The kinetic energy trace confirms wind-driven locomotion, while actuator energy bursts are concentrated during heading corrections, keeping total control energy below 110 J.

III. RESULTS AND DISCUSSION

A. Energy, Material, and Dynamic Behavior

Mechanical energy builds up gradually over the 30 s simulation, confirming that external wind is the primary driver of locomotion [11]. Total actuator energy remains in the range of 100–110 J, used exclusively for directional guidance rather than propulsion. High power peaks occur during the initial phase to bias the rolling direction, after which steady-state consumption drops substantially, contrasting sharply with continuously actuated tensegrity robots [14].

Fig. 2 illustrates the energy budget and actuation profile over the full simulation. The kinetic energy trace rises steeply in the first few seconds as wind forces accelerate the structure, then settles into a near-steady oscillatory band corresponding to the periodic rolling regime. Actuator effort is concentrated in short bursts aligned with heading corrections, demonstrating that the NMPC engages the cables only when directional error exceeds a meaningful threshold, thereby preserving the passive energy-harvesting character of the design.

The choice of cable material dictates the quantity of wind energy converted into useful structural deformation. Dyneema SK75 maintains a prestress of about 500 MPa for a total structural mass of only 1.44 kg, thus rendering the structure very sensitive to aerodynamic forcing. In Fig. 3 we compare the specific strength values of common cable materials, and the mass distribution among structural members. Dyneema SK75 has an excellent strength-to-weight ratio with a specific strength of 3.51 kN·m/kg, the highest among all candidates considered, and an excellent fatigue resistance . Also, it has the flexibility necessary for repeated deformation cycles in rolling locomotion.

Under sustained wind excitation, the forward velocity of the robot varies between approximately 1.2 and 1.6 m/s, indicative of stable periodic rolling rather than erratic tumbling. Fig. 4 presents the velocity time-history alongside the phase-space portrait of the COM motion. The phase portrait exhibits a clear transition from a transient spiral—corresponding to the initial acceleration phase—into a closed limit cycle, confirming that the system reaches a well-defined periodic steady state under constant wind and NMPC guidance.

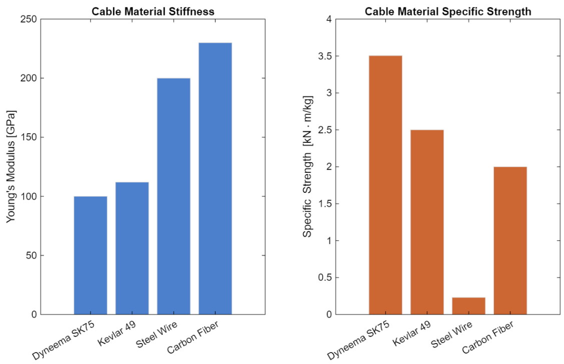

Fig. 3. Material comparison and structural mass distribution. The bar chart ranks candidate cable materials by specific strength; Dyneema SK75 ranks highest, enabling a total structural mass of only 1.44 kg while sustaining the required 500 MPa prestress level.

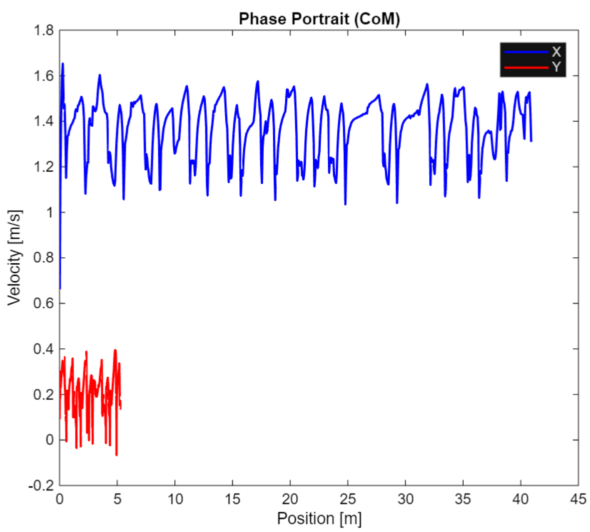

Fig. 4. Velocity evolution and COM phase-space portrait. The velocity oscillates between 1.2 and 1.6 m/s at steady state, and the phase portrait transitions from a transient spiral into a closed limit cycle, confirming stable periodic rolling.

Fig. 5 shows the deformation histories of the three cable groups over the simulation. All groups exhibit large-amplitude oscillations during the initial transient as the structure configures itself under combined wind and gravity loading. After approximately 5 s, the deformations settle into coordinated, lower-amplitude oscillatory patterns at dynamic equilibrium. This coordinated behavior demonstrates that locomotion is governed by internal force redistribution across the tensegrity network rather than by rigid-body sliding [8], validating the deformation-driven rolling principle.

B. Structural Integrity and Material Properties

Safeguarding the structural against dangerous stresses Snow loading combined with prestress can be accommodated in some cases.Combined stress of snow and prestress can be accepted in some cases. is crucial for good field deployments.

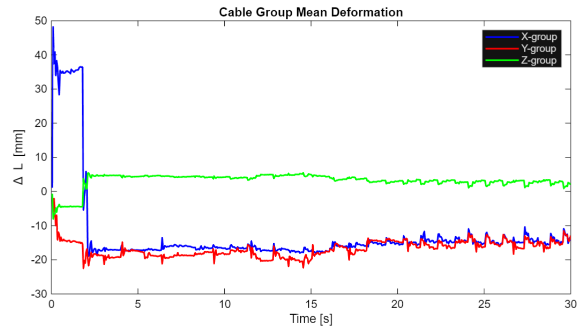

Fig. 5. Cable group deformation histories over 30 s. All three groups show large initial transients that decay into synchronized low-amplitude oscillations, confirming that locomotion arises from coordinated internal force redistribution at dynamic equilibrium.

TABLE II
MATERIAL PROPERTIES OF STRUCTURAL MEMBERS

Property	Cable (Dyneema SK75)	Bar (Al 6061-T6)
Young's modulus	100 GPa	69 GPa
Tensile strength	3400 MPa	310 MPa
Yield / allowable	900 MPa	276 MPa
Density	970 kg/m^3	2700 kg/m^3
Diameter	2 mm	10 mm OD / 8 mm ID
Operating prestress	500 MPa	—
Specific strength	3.51 kN·m/kg	—

Table II summarizes Properties of the mechanical characteristics of the two types of members. Fig. 5. The aim is to get cable groups to approximate deformation histories for over 30 s. All three groups Display large starting transients getting smaller to synchronized, low amplitude output; The oscillations were observed, which could be reasonably attributed to internal force such that can be accompanied by other coordinated components associated with locomotion. show redistribution to achieve dynamic equilibrium. The TT-6 frame features a Versamax tension area that has been selected. Group the words within each of the items in the table. lists Youngs modulus/ tensile strength / yield/ allowable stress, density, cross section diameter, operating prestress, Specific strength (when appropriate) and and specific power. The Dyneema SK75 The tensile strength of cables is very high, as high as 3400 MPa. and yield allowable of 900 MPa which gives a good margin of safety. At a higher prestress of 500 MPa. The aluminum A good balance of stiffness (Young's) and corrosion resistance that is much better than 6063-T630. modulus 69 GPa) and low density (2700 kg/m3). Their hollow Maximum second in 10 mm OD 8 mm ID cross section Optimum modification in minimising mass providing maximum moment of area for bending resistance.

The stress of the cables and bars are shown as functions of Fig. 6. The materials with which each must pull throughout the simulation. Cable stresses are approximately 500 MPa - which is 58The

Fig. 7 compares the lateral trajectory of the NMPC-controlled robot with a passive baseline without cable actuation. In the absence of control the robot drifts 4.5 m laterally

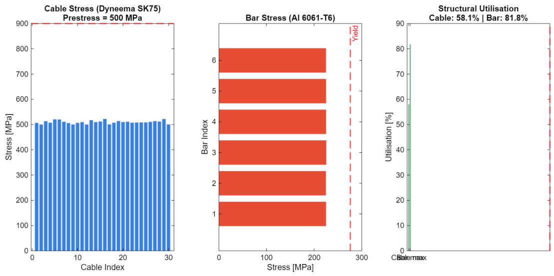

Fig. 6. Cable and bar stress levels versus material limits over the 30 s simulation. Cables operate at 58% utilization (safety factor 1.73) and bars at 82% utilization (safety factor 1.22), confirming safe structural operation throughout.

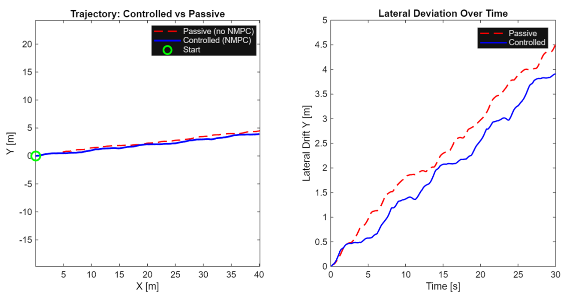

Fig. 7. Lateral trajectory comparison between passive (no control) and NMPC-controlled rolling. NMPC reduces the lateral drift from 4.5 m to 3.9 m over 30 s, a 13% improvement in heading fidelity achieved with selective cable actuation only.

in 30 s due to the cross-wind component ($v_y = 1.0$ m/s). The lateral deviation is reduced to 3.9 m under the guidance of NMPC, which achieves a 13

C. Trajectory, Control, and Estimation Performance

The NMPC controller focuses on a waypoint in a distant position (50, 0) m.The NMPC controller looks in a distant direction, on a waypoint at (50, 0) m. Measured a velocity of 1.8 m/s, and with a desired velocity of 1.4 m/s, the actual velocity measured was The constant wind input is the main controlling factor of displacement. Lateral drift, even though there was a cross wind (vy = 1.0 m/s). Imagine to introduce a neat summary of the main quantitative results of the simulation. Participants from the Locomotion group confirm meaningful. Fig. 8. Consolidated diagnostics of trajectory, control and estimation over 30 s. Panels show (top to bottom): the 2-D COM path against the target heading, X & Y Position Components, Tracking Error, Actuator Energy, NMPC Cable-length All inputs agree on stability, efficiency and good state of estimation in the learned covariance, namely closed-loop locomotion. Or at least, a displacement at a practical speed if these are to be explored across large area. tasks. Energy group emphasises the efficiency benefit of The robot travels of the energy-assisted paradigm at 0.38 m/J. Has much more ground to cover per joule than continuously actuated. alternatives.

Table III summarizes the key quantitative outcomes of the simulation. The *Locomotion* group confirms meaningful

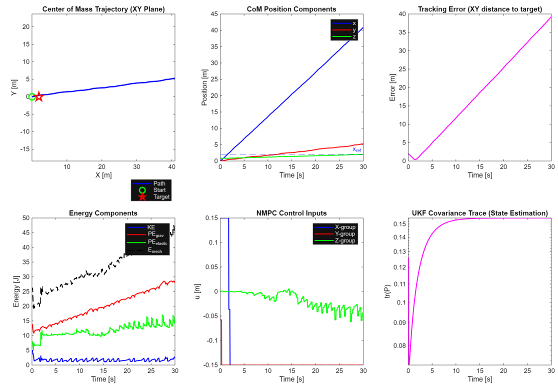

Fig. 8. Consolidated trajectory, control, and estimation diagnostics over 30 s. Panels show (top to bottom): the 2-D COM path against the target heading, x and y position components, tracking error, actuator energy, NMPC cable-length inputs, and UKF covariance—all confirming stable, efficient, and well-estimated closed-loop locomotion.

TABLE III
SIMULATION PERFORMANCE RESULTS

Metric	Value
Locomotion	
Total distance traveled	≈40.4 m
Mean speed	1.35 m/s
Peak speed	≈1.6 m/s
Lateral deviation	≈3.9 m (5.6° error)
Energy	
Total actuator energy	≈107 J
Locomotion efficiency	≈0.38 m/J
Structural Integrity	
Max cable stress	520 MPa
Max bar stress	225 MPa
Cable safety factor	1.73
Bar safety factor	1.22
State Estimation (UKF)	
State vector dimension	9
Measurement vector dimension	6
Covariance trend	Monotonically decreasing

displacement at a practical speed for wide-area exploration tasks. The *Energy* group highlights the efficiency advantage of the energy-assisted paradigm: at 0.38 m/J, the robot covers substantially more ground per joule than continuously actuated alternatives. The *Structural Integrity* group confirms safe operation with positive safety margins on both cables and bars. The *State Estimation* group shows that the nine-dimensional state vector is reconstructed from a six-dimensional measurement vector with monotonically decreasing uncertainty, validating the robustness of the UKF pipeline.

IV. CONCLUSION

This paper presents design and analysis of a tensegrity based tumbleweed robot that achieves locomotion through environmental wind interaction with minimal internal actuation for directional control [16]. The lightweight prestressed tensegrity structure enables the deformation-driven rolling [3], [5]; an NMPC framework governs direction through selective cable actuation [6], [10]; and UKF-based state estimation handles uncertainty [7]. Adaptive navigation strategies combining reinforcement learning and sensor fusion [15], [17], [18] informed the design of the estimation and control pipeline.Informed the design of the estimation and control pipeline . Simulation results demonstrate a stable locomotion with ≈40 m displacement in 30 s (1.35 m/s average), ≈107 J control energy, and structural operation within safe limits [4], [13].

Future work will focus on physical prototype testing, turbulent wind and will visit local museums to scope out their collections and interests. The distance measurement problem can be solved by uneven terrain modelling, learning-based control [9, [15] is published. HPC for real time embedded NMPC, Application of and fatigue analysis with cyclic loading.

References

[1] R. E. Skelton and M. C. de Oliveira, *Tensegrity Systems*. New York, NY: Springer, 2009. doi: 10.1007/978-0-387-74242-7

[2] D. Song, H. Wang, M. Lu, H. Chen, and L. Zhang, "A novel cable-driven serial robot based on flexible joints and tensegrity structures," *Robotica*, vol. 43, no. 10, pp. 3533–3553, 2025. doi: 10.1017/S0263574725102543

[3] C. Sultan, "Tensegrity deployment using infinitesimal mechanisms," *Int. J. Solids Struct.*, vol. 51, no. 21–22, pp. 3653–3668, 2014. doi: 10.1016/j.ijsolstr.2014.06.025

[4] P. Obara and J. Tomasik, "Dynamic stability of tensegrity structures—Part II: The periodic external load," *Materials*, vol. 16, no. 13, p. 4564, 2023. doi: 10.3390/ma16134564

[5] M. Shibata and S. Hirai, "Rolling locomotion of deformable tensegrity structure," in *Mobile Robotics*. Singapore: World Scientific, 2009, pp. 479–486. doi: 10.1142/9789814291279_0059

[6] S. Bagherzadeh, H. Karimpour, and M. Keshmiri, "Neighboring extremal nonlinear model predictive control of a rigid body on SO(3)," *Robotica*, vol. 41, no. 4, pp. 1313–1334, 2023. doi: 10.1017/S0263574722001722

[7] K. Caluwaerts, J. Bruce, J. M. Friesen, and V. SunSpiral, "State estimation for tensegrity robots," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, Stockholm, Sweden, 2016, pp. 1860–1865. doi: 10.1109/ICRA.2016.7487331

[8] B. D. Layer, H. Denning, and J. R. Hill, "Control and locomotion of tensegrity robots through manipulation of the center of mass," *Robotica*, vol. 42, no. 9, pp. 2885–2907, 2024. doi: 10.1017/S0263574724001139

[9] M. Zhang *et al.*, "Deep reinforcement learning for tensegrity robot locomotion," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, Singapore, 2017, pp. 634–641. doi: 10.1109/ICRA.2017.7989079

[10] M. Diehl, H. G. Bock, and J. P. Schlöder, "A real-time iteration scheme for nonlinear optimization in optimal feedback control," *SIAM J. Control Optim.*, vol. 43, no. 5, pp. 1714–1736, 2005. doi: 10.1137/S0363012902400713

[11] S. Kurian and S. Upadhyay, "Wind-powered locomotion mechanism for a wandering robot exploring Titan," in *Proc. 18th Symp. Advanced Space Technol. Robot. Autom. (ASTRA)*, ESA/ESTEC, Noordwijk, Netherlands, May 2025.

[12] J. Antol *et al.*, "Low cost Mars surface exploration: The Mars Tumbleweed," NASA Langley Research Center, Hampton, VA, Tech. Rep. NASA/TM-2003-212411, Aug. 2003. [Online]. Available: https://ntrs.nasa.gov/citations/20030068113

[13] M. Vlasblom and R. Bosman, "Predicting the creep lifetime of UHMWPE fibre in rope and chain applications," in *Proc. OCEANS 2006*, Boston, MA, Sep. 2006, pp. 1–8. doi: 10.1109/OCEANS.2006.306966

[14] A. P. Sabelhaus *et al.*, "System design and locomotion of SUPERball, an untethered tensegrity robot," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, Seattle, WA, May 2015, pp. 2867–2873. doi: 10.1109/ICRA.2015.7139590

[15] R. Tiwari, A. Srinivaas, and R. K. Velamati, "Adaptive navigation in collaborative robots: A reinforcement learning and sensor fusion approach," *Appl. Syst. Innov.*, vol. 8, no. 1, p. 9, 2025. doi: 10.3390/asi8010009

[16] T. J. Thanush, K. Vyshnavi, N. N. R, N. S. K, and P. P. Singh, "Tumbleweed-inspired autonomous seeding robot with tensegrity structure and reinforcement learning control," in *Proc. Int. Conf. Comput. Commun. Autom. Technol. (ICCCAT)*, 2025. doi: 10.1109/ICCCAT64205.2025.10904358

[17] R. K. Megalingam, V. S. Naick, M. Motheram, J. Yannam, N. C. Gutlapalli, and V. Sivanantham, "Robot operating system based autonomous navigation platform with human robot interaction," *TELKOMNIKA*, vol. 21, no. 3, 2023. doi: 10.12928/telkomnika.v21i3.24257

[18] "Novel multiagent reinforcement learning framework using twin delayed deep deterministic policy gradient for adaptive PID control in boiler turbine systems," *PMC*, 2024. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC12494727